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Products and coproducts

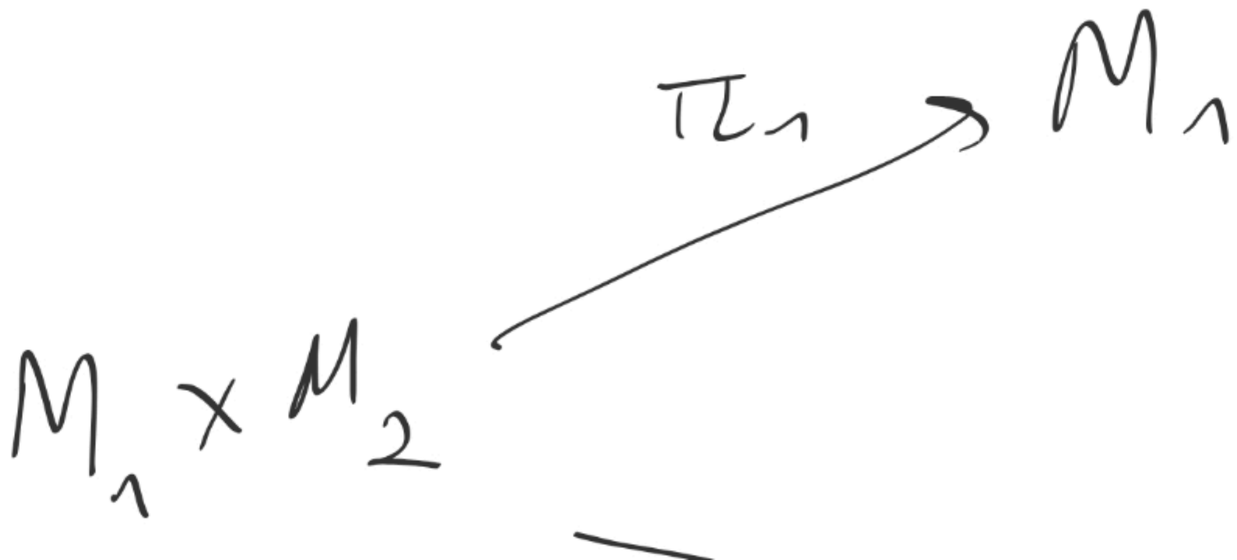
A, B sets

$\mathrm{Hom}_{\mathcal{C}}(A, B) =$ the set of all mappings $A \to B$

G, H groups

$\mathrm{Hom}_{\mathcal{G}}(G, H) =$ the set of all morphisms $G \to H$.

M_1, M_2 sets





$$E \xrightarrow{f} M_1 \times M_2$$

$$\rightsquigarrow \quad \pi_1 \circ f : E \longrightarrow M_1$$

$$\pi_2 \circ f : E \longrightarrow M_2$$

$$f \in \mathrm{Hom}_{\mathcal{E}}(E, M_1 \times M_2)$$

$$\rightsquigarrow \quad \pi_1 \circ f \in \mathrm{Hom}_{\mathcal{E}}(E, M_1)$$

$$\pi_2 \circ f \in \mathrm{Hom}_{\mathcal{E}}(E, M_2).$$

Get a mapping

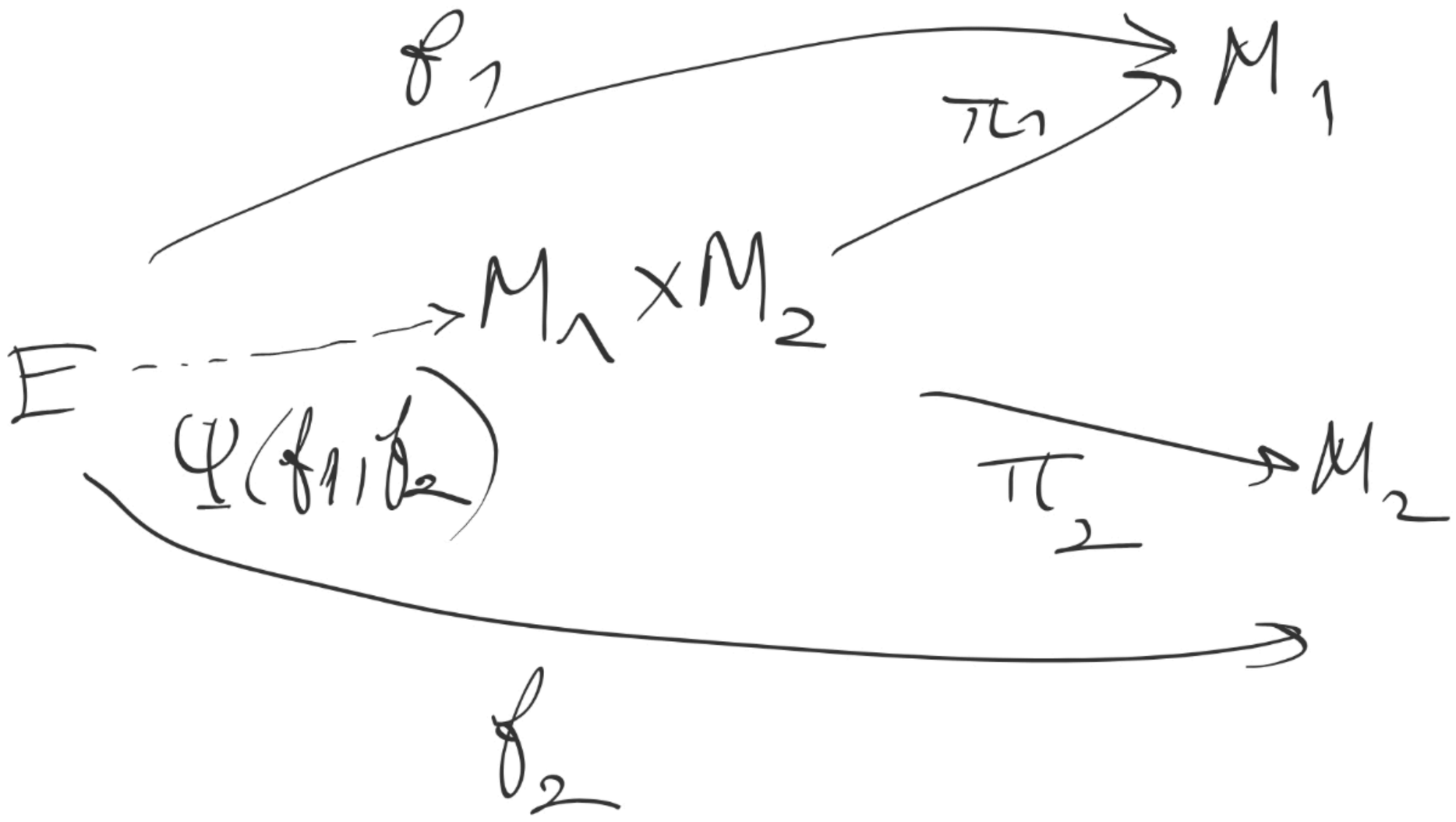
$$\mathrm{Hom}_{\mathcal{E}}(E, M_1 \times M_2) \xrightarrow{\Phi} \mathrm{Hom}_{\mathcal{E}}(E, M_1) \times \mathrm{Hom}_{\mathcal{E}}(E, M_2)$$

$$f \longmapsto (\pi_1 \circ f, \pi_2 \circ f)$$

Prop Φ is a bijection.

Proof Given $(f_1, f_2) \in \underset{\mathcal{E}}{\mathrm{Hom}}(E, M_1) \times \underset{\mathcal{E}}{\mathrm{Hom}}(E, M_2)$

let $\Psi(f_1, f_2) \in \underset{\mathcal{E}}{\mathrm{Hom}}(E, M_1 \times M_2)$ be the unique mapping making these two triangles commute



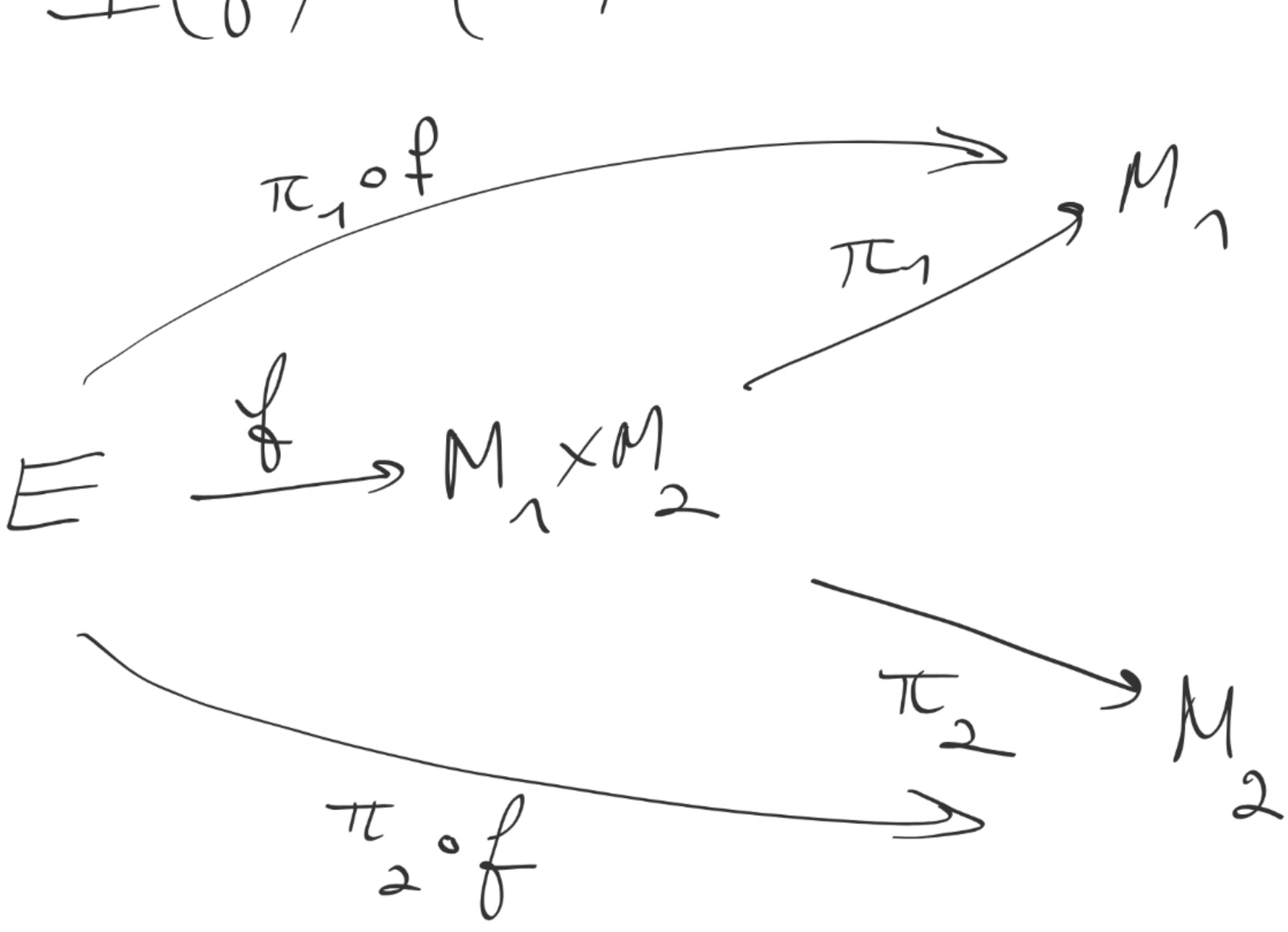
? $\Psi \circ \Phi(f) = f$ for every $f \in \mathrm{Hom}(E, M_1 \times M_2)$?

? $\Phi \circ \Psi(f_1, f_2) = (f_1, f_2)$

for every $(f_1, f_2) \in Hom(E, M_1) \times Hom(E, M_2)$

$$f : E \longrightarrow M_1 \times M_2$$

$$\Phi(f) = (\pi_1 \circ f, \pi_2 \circ f)$$

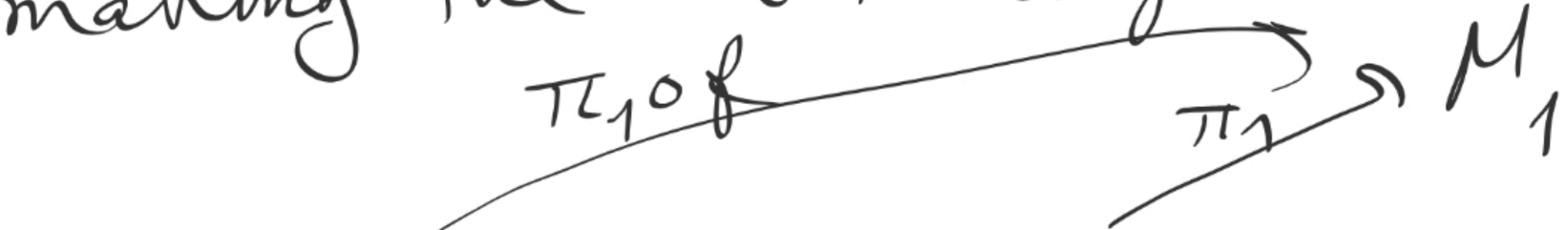


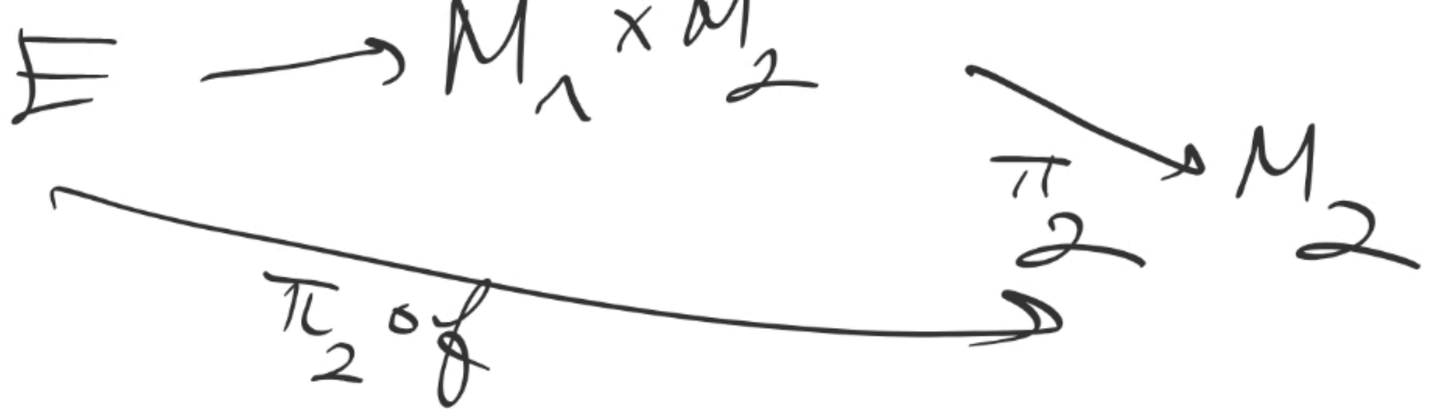
$$\Psi(\pi_1 \circ f, \pi_2 \circ f) = \zeta$$

by def, this is the unique mapping

$$\zeta : E_1 \longrightarrow M_1 \times M_2$$

making the two triangles

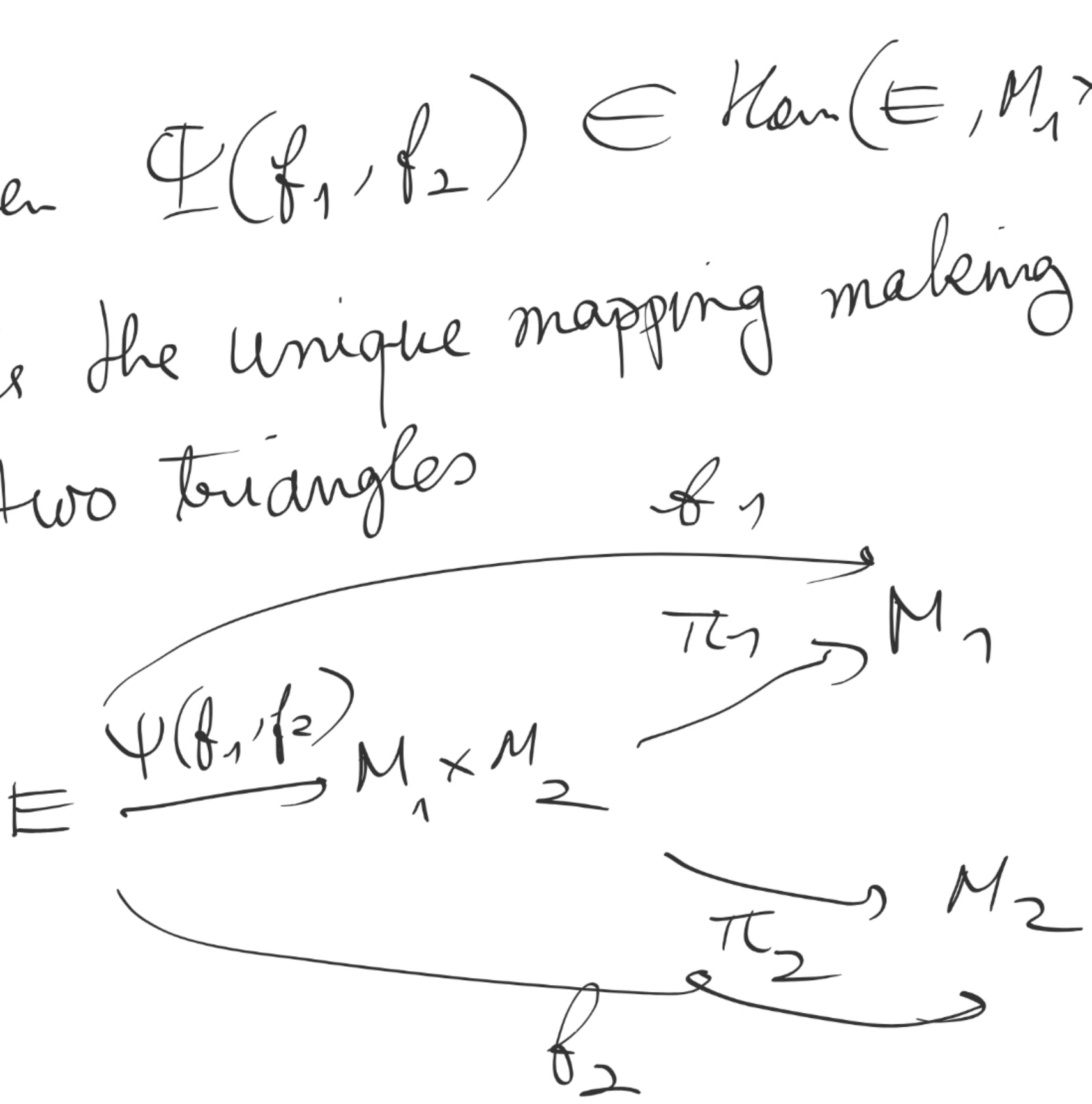




Let $(f_1, f_2) \in \mathrm{Hom}(E, M_1) \times \mathrm{Hom}(E, M_2)$

Then $\Psi(f_1, f_2) \in \mathrm{Hom}(E, M_1 \times M_2)$

is the unique mapping making the two triangles f_1



$\Phi(\Psi(f_1, f_2)) = ?$

$= (\pi_1 \circ \Psi(f_1, f_2), \pi_2 \circ \Psi(f_1, f_2))$

$\Rightarrow (f_1, f_2)$,

Conclusion The mapping

$$\overline{\Phi} : \mathrm{Hom}(E, M_1 \times M_2) \to \mathrm{Hom}(E, M_1) \times \mathrm{Hom}(E, M_2)$$

$$\overline{\Phi}(f) = (\pi_1 \circ f, \pi_2 \circ f)$$

□

is a bijection.

Generalisation

$$\mathrm{Hom}\Big(E, \prod_{\lambda \in L} M_\lambda\Big) \xrightarrow{\overline{\Phi}} \prod_{\lambda \in L} \mathrm{Hom}(E, M_\lambda)$$

$$\overline{\Phi}(f) = (\pi_\lambda \circ f)_{\lambda \in L}$$

is a bijection.

Similarly,

$$\mathrm{Hom}\Big(\coprod_{\lambda\in L} M_\lambda, E\Big) \longrightarrow \prod_{\lambda\in L} \mathrm{Hom}(M_\lambda, E)$$

$$f \longmapsto (f\circ i_\lambda)_{\lambda\in L}$$

is a bijection with reciprocal

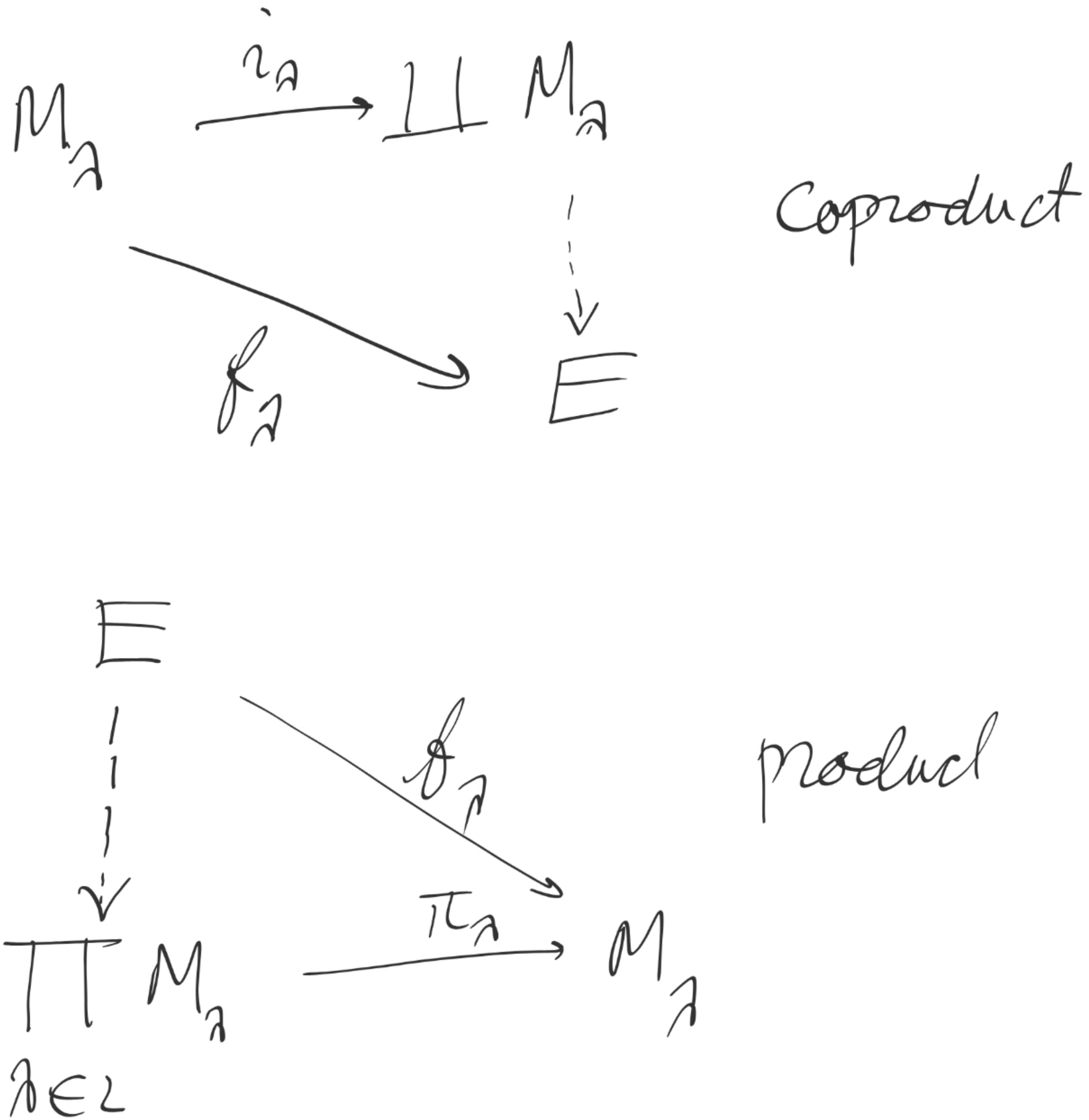
$$\psi\big((f_\lambda)_{\lambda\in L}\big) \longmapsfrom (f_\lambda)_{\lambda\in L}$$

where $\psi\big((f_\lambda)_{\lambda\in L}\big)$ is the unique mapping $\coprod_{\lambda\in L} M_\lambda \longrightarrow E$ such that

$$\psi\big((f_\lambda)_{\lambda\in L}\big)\circ i_\lambda = f_\lambda$$

for every $\lambda\in L$.

Family of sets $(M_\lambda)_{\lambda\in L}$


M_λ
i_λ
$\coprod M_\lambda$
f_λ
E
Coproduct
E
f_λ
product
$\prod M_\lambda$
π_λ
M_λ
$\lambda \in \Lambda$